

Supplementary Information

Protocol Sensitivity in Remaining-Useful-Life Model Comparisons:
Retrospective C-MAPSS Analysis and a Pre-Outcome-Frozen
Battery Study

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Evidence identity. The source release is identified by the annotated GitHub tag v5.5.0; its resolved tag, commit, and tree hashes are recorded in the external submission identity record. The v5.5.0 evidence archive is publicly available as a version-named file in the OSF project at <https://osf.io/download/6ab6a6735205d2cb18154961/>. Its SHA-256 is A390CA0216E9CA1238420CA3F76AAFD536A18B07D1B0FF2631FAA7FDF50566B. This is a public project file, not a new OSF registration or immutable version DOI. The final PDF hashes are listed in the same identity record; the OSF archive hash identifies the evidence ZIP, not the compiled PDFs.

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S1 Scope, direction convention, and evidence vocabulary

Every difference is first named configuration minus second named configuration. Negative values favor the first configuration for RMSE, NASA/engine, late-prediction ratios, and normalized maintenance-decision cost. “Completed levels” are fixed study objects, not draws from a defined population. “Diagnostic” means that an output describes the completed design without calibrated confirmatory coverage or a prospective decision role. Table S1 defines the evidence vocabulary used in this concise Supplement.

Table S1: Evidence vocabulary used in both documents.

Term			Operational meaning	Prohibited interpretation
Fixed-task estimate			Paired mean and sign pattern over five completed coupled levels	Population effect, confirmatory test, or architecture main effect
Sensitivity			Observed direction, magnitude, or support over declared post-result choices	Prevalence over all defensible protocols
Diagnostic envelope			Resampling visualization of the completed design	Calibrated 95% confidence interval
Prospectively	frozen		Battery protocol timestamped before training and test-label evaluation	Independent-team replication or model superiority
Static	RUL	trigger	Service/no-service mapping under declared RUL trigger threshold and cost grids	Optimized policy, monetary utility, or system reliability

S2 Traceability scaffold and neighboring reporting frameworks

The six-part traceability scaffold is supporting infrastructure for the protocol-sensitivity analysis, not a validated intervention or replacement for existing reporting frameworks. REFORMS is a consensus checklist for ML-based science [1]; Model Cards document model-level intended use and evaluation [2]; Eval Factsheets organize evaluation documentation [3]; and Evaluation Cards provide an interpretive evaluation-reporting layer with large-scale deployment evidence [4]. Chen et al.’s Evidence-Ledger Adjudication links claims to evidence packets and support relations for AI-assisted writing [5]; its unit is a claim-verification packet, not an RUL benchmark comparison or protocol-sensitivity record. The present study has no independent comparative-effect result for the scaffold. Table S2 maps its fields to neighboring reporting frameworks without claiming superiority.

Figure S1 shows the scaffold, and Table S3 defines its six operational fields.

Table S2: Construct map. “Explicit” describes a named field, not evidence that the format is superior.

Framework	Primary object	Design timing	Provenance	Claim/wording constraint
Six-part traceability scaffold	Individual benchmark claim	Explicit field	Explicit artifact field	Explicit permitted-wording field; efficacy untested
REFORMS	ML-based scientific study	Study-design recommendations	Computational-reproducibility module	Scope, metric, and limitation guidance
Evaluation Cards	Evaluation record and reader interpretation	Timing-relevant metadata	Central interpretive signal	Reader modes and comparability signals
Eval Factsheets	Evaluation documentation object	Context and procedure fields	Documentation-centered	Interpretive documentation across five dimensions
Evidence-Ledger Adjudication	Claim/evidence packet	Per-claim adjudication	Evidence packet and support relation	Routes unsupported, contradicted, or mixed claims for review
Model Cards	Trained model report	Training/evaluation procedure	Model/version record	Intended use, limitations, and disaggregated evaluation
Preregistration	Study plan	Central purpose	Immutable pre-outcome plan	Constrains post-result interpretation indirectly
Multiverse reporting	Analysis-choice distribution	Prospective or retrospective	Implementation-dependent	Exposes completed specification behavior

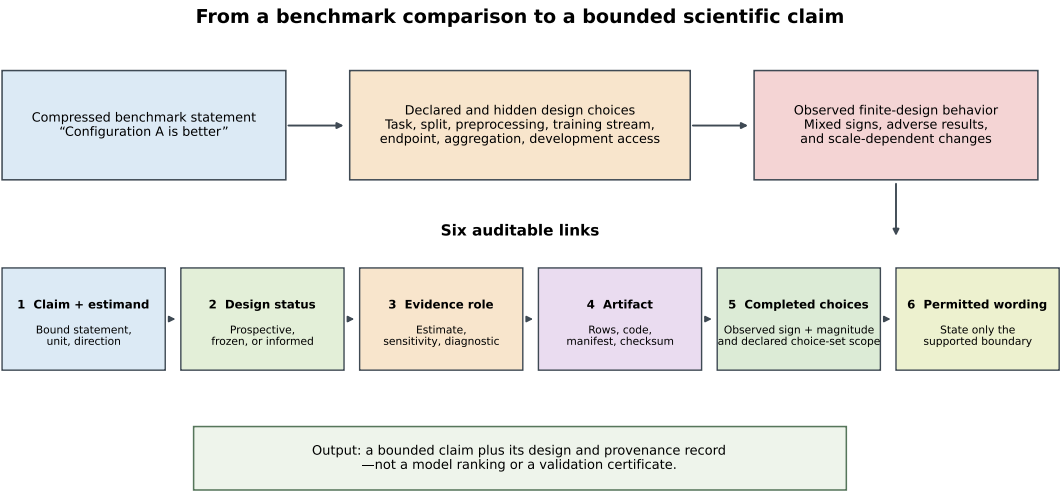


Figure S1: Six-part traceability scaffold used as supporting documentation for the protocol-sensitivity analyses. The scaffold records comparison unit, design timing, evidence support, artifacts, completed sensitivities, and warranted wording; it does not upgrade the underlying evidence.

Table S4: Task-specific finite-design effects for OCM-Asym minus RAST-GRU. Ranges and signs summarize five completed coupled levels; they are not confidence intervals or population-inference statistics. RMSE effects are also shown as a percentage of the 125-cycle cap, and LPR effects as percentage-point changes; neither scale is an engineering equivalence margin. N_{30} is the LPR denominator.

Task	Metric	Mean effect (descriptive scale)	Observed range	- / + / 0	N_{test}	N_{30}
FD001	RMSE (cycles)	-0.806 (-0.64%)	[-2.453, +0.159]	3/2/0	100	25
FD001	NASA / engine	-0.537	[-1.305, +0.216]	4/1/0	100	25
FD001	LPR@30	-0.040 (-4.0 pp)	[-0.320, +0.160]	3/2/0	100	25
FD002	RMSE (cycles)	-0.260 (-0.21%)	[-0.891, +0.975]	4/1/0	259	61
FD002	NASA / engine	-0.619	[-1.137, +0.187]	4/1/0	259	61
FD002	LPR@30	-0.036 (-3.6 pp)	[-0.213, +0.082]	2/3/0	259	61
FD003	RMSE (cycles)	+0.664 (+0.53%)	[-0.422, +2.261]	2/3/0	100	20
FD003	NASA / engine	+0.982	[-0.133, +3.064]	1/4/0	100	20
FD003	LPR@30	+0.040 (+4.0 pp)	[-0.100, +0.200]	2/2/1	100	20
FD004	RMSE (cycles)	+0.893 (+0.71%)	[-1.143, +2.461]	1/4/0	248	53
FD004	NASA / engine	-1.241	[-9.647, +1.199]	1/4/0	248	53
FD004	LPR@30	-0.117 (-11.7 pp)	[-0.245, -0.038]	5/0/0	248	53

Table S3: Operational fields in the six-part traceability scaffold.

Field	Required content
Comparison and unit	Quantity, direction, aggregation rule, and independent evaluation unit.
Design timing	When consequential choices were fixed and whether outcomes had already been inspected.
Evidence support	What the completed design can support without upgrading diagnostics to population inference.
Artifacts	Resolvable data rows, code, manifests, stable identifiers, and hashes needed to inspect the claim.
Completed sensitivities	Adverse, null, mixed, and reversing outcomes, reported together with their scale.
Claim wording	The strongest sentence warranted by the preceding five fields, without unsupported superiority or practical-benefit claims.

S3 C-MAPSS finite-design evidence

FD004 is development-informed because it supplied the original preference-development split. The statistical unit for endpoint comparisons is the matched official test engine; the five composite levels jointly change split, initialization, shuffle, and augmentation.

The archived raw files used for FD004 contain 249 unique training-unit identifiers and 248 test-unit identifiers. Their SHA-256 hashes are 27ef6160b6a1dcb2613a88de9c239f763b223f02cdc41dc5cdedc5dc189b6218 (train_FD004.txt) and 1dc675fff0624bac10786927c6715b37d1297657137400d2b1a3138d777a3ba5 (test_FD004.txt); both match the corresponding files inside the archived CMAPSSData.zip (SHA-256 74bef434a34db25c7bf72e668ea4cd52afe5f2cf8e44367c55a82bfd91a5a34f). NASA’s current Open Data landing page reports 248 training and 249 test trajectories [6]. The experiments use the archived local files identified by these hashes.

Table S5: Integer support for the targeted LPR@30 endpoint. Entries are observed late-event-count ranges across five coupled levels, shown as OCM-Asym / RAST-GRU.

Task	Eligible N_{30}	$\epsilon = 0$ events	$\epsilon = 5$ events
FD001	25	0–4 / 0–8	0–0 / 0–0
FD002	61	3–8 / 1–16	0–0 / 0–2
FD003	20	1–6 / 1–5	0–1 / 0–1
FD004	53	7–17 / 15–23	0–4 / 3–7

S4 Full split–stream crossing

The 400-run extension pairs five split levels with ten training streams for each of four tasks and two complete configurations. Each task therefore has 50 matched split–stream cells and one estimate per cell. The formal execution began before the plan and Amendment 01 were frozen on 22 September 2026. The lock is pre-aggregation, not pre-execution preregistration. The complete aggregate was generated after the 400-row manifest was checked and hashed. All rows used validation last-risk-score checkpoint selection, an 80-epoch cap, and patience 12. The 400 runs stopped by the declared early-stopping rule between epochs 13 and 61; each history reached the patience condition and retained its selected checkpoint.

The heatmaps below show cell-level contrasts as OCM-Asym minus RAST-GRU. Negative cells favor OCM-Asym for the displayed error endpoint. Rows are split levels and columns are training streams. Each panel shows the finite design, not a sample from a split or stream population.

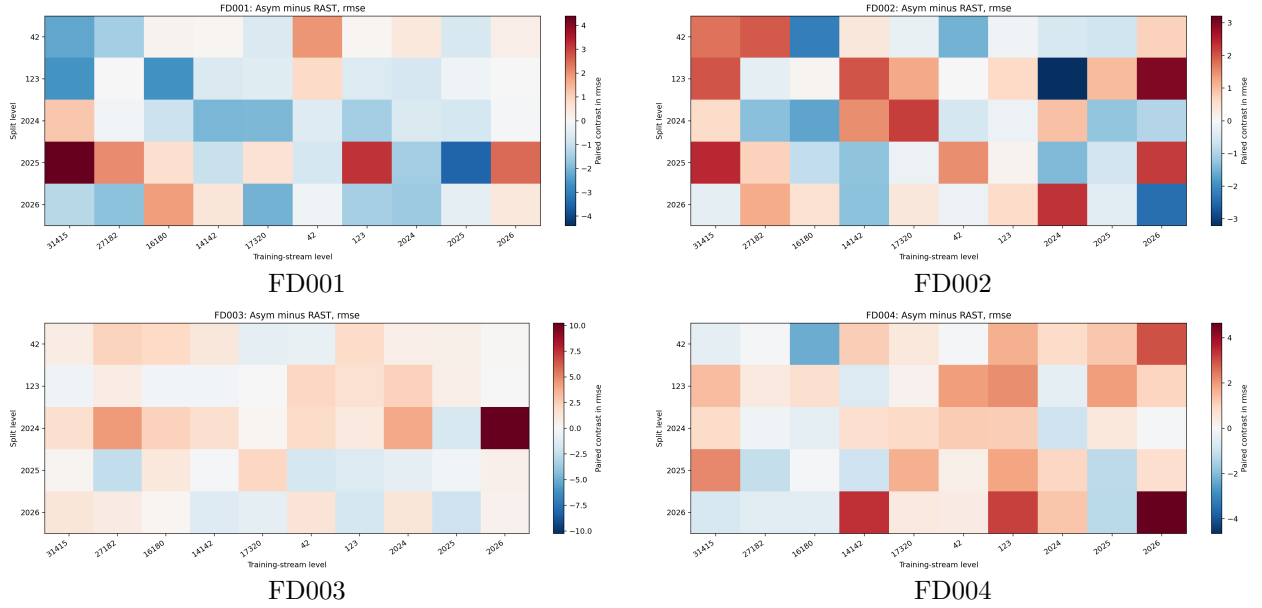


Figure S2: RMSE contrasts across the 5-by-10 crossing.

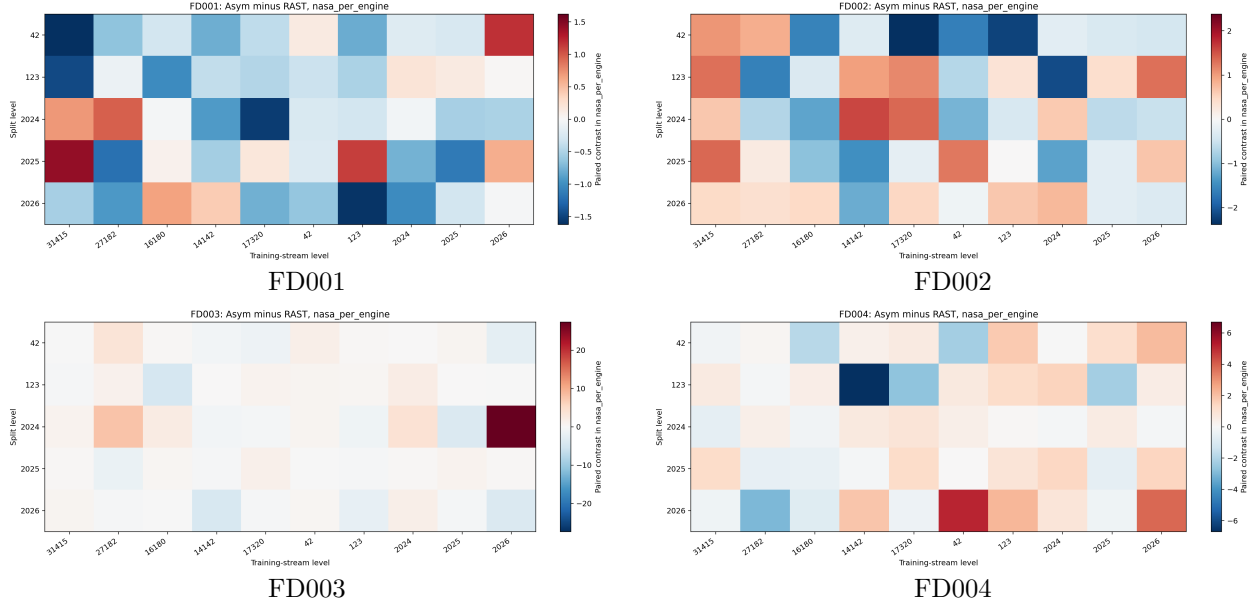


Figure S3: NASA/engine contrasts across the 5-by-10 crossing.

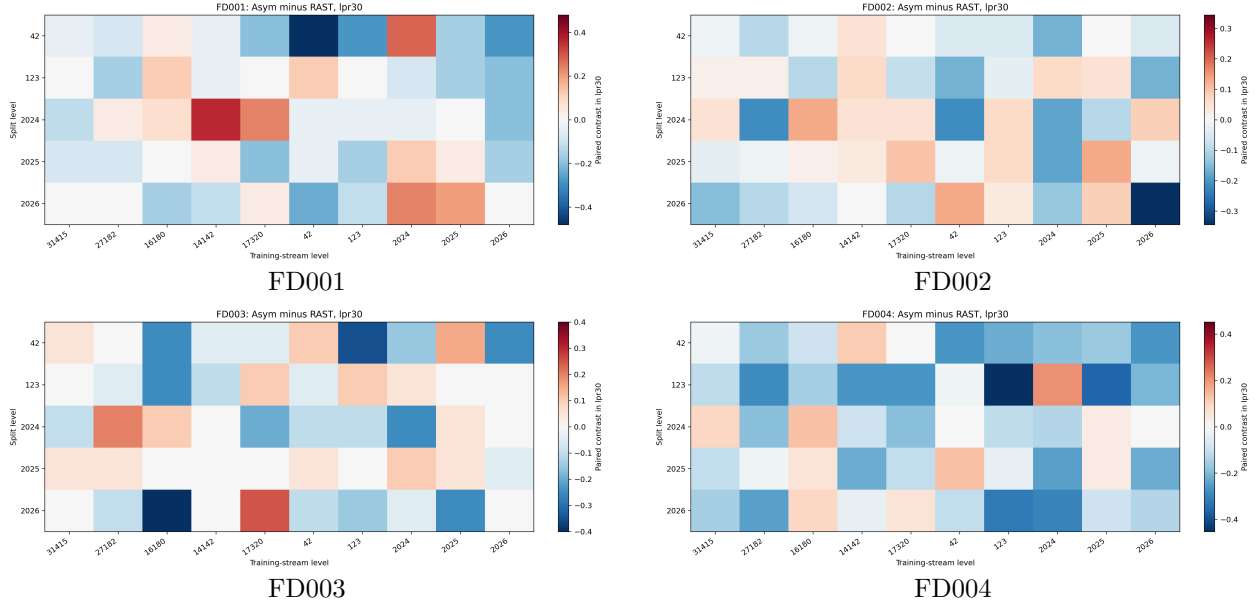


Figure S4: LPR@30 contrasts across the 5-by-10 crossing.

Table S6: Finite-design OCM-Asym minus RAST-GRU contrasts across the full split-stream crossing. Each task has 50 matched cells. Sign counts are negative/tied/positive.

Task	RMSE mean	RMSE -/0/+	NASA/engine mean	NASA -/0/+	LPR@30 mean	LPR -/0/+
FD001	-0.318	33/0/17	-0.307	36/0/14	-0.041	28/8/14
FD002	+0.171	24/0/26	-0.124	27/0/23	-0.031	27/3/20
FD003	+0.719	18/0/32	+0.618	22/0/28	-0.039	22/13/15
FD004	+0.653	18/0/32	+0.203	21/0/29	-0.113	37/3/10

Table S7: Integer support for LPR@30 in the full split-stream crossing. Event ranges are the minimum–maximum count over the 50 cells per task.

Task	Matched cells	Eligible engines per cell	RAST/OCM late-event count range
FD001	50	25	0–14 / 0–9
FD002	50	61	0–23 / 0–11
FD003	50	20	0–12 / 0–8
FD004	50	53	5–30 / 3–23

Table S8: Pairwise finite-design rank support over the full split-stream crossing. Counts are OCM-Asym lower metric / tied / RAST-GRU lower metric among 50 matched cells.

Task	Endpoint	OCM lower	Tie	RAST lower
FD001	RMSE	33	0	17
FD001	NASA/engine	36	0	14
FD001	LPR@30	28	8	14
FD001	MAE	26	0	24
FD001	SLPR@30,5	8	35	7
FD001	SLPR@30,10	0	50	0
FD002	RMSE	24	0	26
FD002	NASA/engine	27	0	23
FD002	LPR@30	27	3	20
FD002	MAE	20	0	30
FD002	SLPR@30,5	6	35	9
FD002	SLPR@30,10	0	49	1
FD003	RMSE	18	0	32
FD003	NASA/engine	22	0	28
FD003	LPR@30	22	13	15
FD003	MAE	13	0	37
FD003	SLPR@30,5	14	27	9
FD003	SLPR@30,10	2	48	0
FD004	RMSE	18	0	32
FD004	NASA/engine	21	0	29
FD004	LPR@30	37	3	10
FD004	MAE	12	0	38
FD004	SLPR@30,5	41	3	6
FD004	SLPR@30,10	21	25	4

S5 Endpoint roles and development relations

The main text distinguishes target-monitored endpoints from independent downstream validation signals. Table S9 records how the reported quantities relate to training and checkpoint selection. Reporting an endpoint on held-out test engines does not make it independent of model development or checkpoint selection.

Table S10: Transparent candidate magnitude-band sensitivity.

Domain	Metric	Margin	Mean contrast	Within	Negative beyond	Positive beyond
C-MAPSS	RMSE	0.5	+0.1227	5/20	7	8
C-MAPSS	RMSE	1	+0.1227	14/20	3	3
C-MAPSS	RMSE	2	+0.1227	17/20	1	2
C-MAPSS	LPR30	0.01	-0.0383	1/20	12	7
C-MAPSS	LPR30	0.025	-0.0383	2/20	12	6
C-MAPSS	LPR30	0.05	-0.0383	8/20	8	4
NASA battery	RMSE	0.5	+2.5455	1/5	0	4
NASA battery	RMSE	1	+2.5455	2/5	0	3
NASA battery	RMSE	2	+2.5455	2/5	0	3
NASA battery	LPR	0.01	-0.0777	2/5	2	1
NASA battery	LPR	0.025	-0.0777	2/5	2	1
NASA battery	LPR	0.05	-0.0777	2/5	2	1

Notes: Negative/positive are first-minus-second contrasts beyond the displayed symmetric margin. Margins were introduced after outcome access and are not application-validated smallest effects of interest. The table separates sign sensitivity from magnitude sensitivity; it is not a formal equivalence test.

Table S9: Endpoint-role map used to constrain interpretation.

Endpoint	Training-objective relation	Checkpoint relation	Permitted interpretation
RMSE	Aligned with point-error accuracy	Direct component of S_{val}	Target-monitored accuracy, not an independent endpoint
LPR@30, SLPR@30,10	Thirty-cycle life weighting is shared; OCM adds a smooth near-failure positive-error term	Direct components of S_{val}	Target-conditioned near-failure-overestimation endpoints
NASA/engine	Not directly optimized	Not in S_{val}	Out-of-target asymmetric-error sensitivity
Alternative τ or positive ϵ	Related to, but not identical to, the 30-cycle target	Not in S_{val}	Threshold or tolerance sensitivity
ZIMLE, CMLE, CVaR, A_κ	Objective-adjacent error summaries	Not in S_{val}	Descriptive magnitude/tail sensitivity, not maintenance utility

S6 Magnitude and maintenance-decision sensitivities

Candidate magnitude bands were introduced after outcome access and are not stakeholder-validated smallest effects of interest. The static action is $a_i(h) = \mathbb{I}(\hat{y}_i \leq h)$ and the reference due-state is $z_i(h) = \mathbb{I}(y_i \leq h)$. Normalized loss is $\rho z_i(h)[1 - a_i(h)] + [1 - z_i(h)]a_i(h)$. It excludes inspection duration, replacement lead time, downtime, failure probability, and physical consequence.

S6.1 Observed-prefix policy grid

The policy grid uses only current and prior predictions to trigger actions. True RUL enters only cost scoring. Thresholds are 10, 20, 30, and 40 cycles; lead times are 0, 5, and 10 cycles; persistence requirements are 1, 2, and 3 windows; corrective-cost ratios are 5, 10, and 20. The fixed-age comparator triggers at cycle 100, and an always-corrective comparator is also retained. A trigger fixed before the observation cutoff can be scored after the cutoff. The analysis never infers a new post-cutoff trigger.

Unresolved engines remain in the denominator for resolution and unresolved rates. Conditional

Table S11: Exploratory static maintenance-triage sensitivity for OCM-Asym minus RAST-GRU.

RUL trigger threshold h	Ratio ρ	$\Delta C/\text{engine}$	Cost $-/0/+$	Action disagree.	Δ false-late
10	2	+0.0017	8/4/8	0.0228	-0.0011
10	5	-0.0017	10/3/7	0.0228	-0.0011
10	10	-0.0073	11/3/6	0.0228	-0.0011
20	2	-0.0030	11/3/6	0.0178	-0.0014
20	5	-0.0072	11/2/7	0.0178	-0.0014
20	10	-0.0142	11/2/7	0.0178	-0.0014
30	2	+0.0033	4/8/8	0.0124	+0.0002
30	5	+0.0039	6/7/7	0.0124	+0.0002
30	10	+0.0048	7/6/7	0.0124	+0.0002

Notes: Service is triggered when predicted RUL is no greater than the declared RUL trigger threshold. Cost equals C_L for a deferred action when true RUL is within the threshold and C_E for premature service otherwise. Each summary gives an equal-weight mean over four tasks and five composite levels. Negative differences favor OCM-Asym. This is a post-result static decision sensitivity, not an optimized fleet policy, failure-probability model, or field utility estimate.

costs are means among resolved engines only. Paired OCM–RAST cost uses engines resolved under both configurations. Reference-only, comparison-only, both-resolved, and neither-resolved counts are kept for each grid cell. Table S12 summarizes paired setting-level cost differences; its setting counts are not independent observations and supports vary across settings.

Table S12: Observed-prefix paired cost contrasts over the declared dynamic policy grid. Each task has 5,400 dependent policy/split/stream settings. Cost delta is OCM-Asym minus RAST-GRU among jointly resolved engines; negative values favor OCM-Asym.

Task	Settings $-/0/+$	Median cost delta	Observed range	Paired support min/median/max
FD001	2262/36/3102	+0.00519	[−18.880, 5.397]	1/22.5/33
FD002	1707/9/3684	+0.01243	[−6.263, 3.493]	18/60.5/92
FD003	2079/27/3294	+0.00969	[−18.885, 12.588]	2/18/30
FD004	1623/9/3768	+0.01406	[−8.797, 4.183]	7/51.5/83

Table S13: Ranges across the 5,400 declared dynamic-policy settings per task for resolution rate and conditional mean cost. Costs are calculated among resolved engines; ranges are not confidence intervals.

Task	Configuration	Resolution-rate range	Mean cost among resolved range
FD001	RAST-GRU	0.010–0.360	1.097–20.000
	OCM-Asym	0.060–0.360	1.117–5.851
FD002	RAST-GRU	0.070–0.371	1.134–7.439
	OCM-Asym	0.104–0.390	1.145–4.927
FD003	RAST-GRU	0.020–0.360	1.085–20.000
	OCM-Asym	0.040–0.400	1.105–13.708
FD004	RAST-GRU	0.028–0.363	1.274–10.811
	OCM-Asym	0.036–0.359	1.267–10.270

S7 Retrospective protocol sensitivities retained in the archive

The concise document reports the main interpretation-changing checks; the complete row-level registries and all secondary tables remain machine readable. None is confirmatory. Table S14 provides a compact dashboard of those completed sensitivity families.

Table S14: Compact sensitivity dashboard. Directions are OCM-Asym minus RAST-GRU.

Family	Completed support	Interpretation
Fixed-split streams	Ten optimization streams per task at one split	LPR mean is lower on all four tasks, but split variation is absent
Selected split-stream grid	Negative LPR directions: 4/9, 5/9, 4/9, 7/9 on FD001–FD004	Result-informed finite grid; no split or stream population inference
FD004 preprocessing	Window 50 reverses mean RMSE and LPR; 21 sensors reverses mean LPR	Representation, capacity, and optimization exposure change together
LOFO	Random missingness is adverse; clean correction reverses some block/drift summaries	Conditional on normalized-coordinate augmentation construction
N-CMAPSS proxy	Same three test units under thinning/window controls	Representation dependence, not external validation

S7.1 N-CMAPSS representation diagnostic

The N-CMAPSS DS02-006 analysis is deliberately narrower than an external validation study [7]. It keeps every 100th record, forms overlapping length-50 windows, and reuses the same three official held-out test units while selected development, sampling, and window choices are changed. Because the representation is heavily thinned and the test units are unchanged, the analysis is used only to show representation dependence. It is not treated as field run-to-failure evidence, a CruiseBench reproduction, or an independent benchmark validation.

Under the declared validation-only condition-count rule, $K = 4$ has the lowest validation risk (0.189), but retrospective test behavior is not uniformly favorable under the reused three test units. The result is therefore retained as a computational representation diagnostic rather than an external RUL validation claim.

S7.2 P10 comparator provenance

The archived P10 extension is not part of the primary comparison. It contains 400 C-MAPSS runs, paired as 200 task-split-stream cells, together with grouped battery and baseline runs. Its C-MAPSS labels are `rast_gru_v2` and `rast_gru`; the source record does not establish that `rast_gru_v2` is OCM-Asym. In addition, P10 selects checkpoints with `val_last_rmse`, whereas the focal C-MAPSS protocol uses `val_last_risk_score`. The two result families therefore remain separate. P10 is retained here as provenance, not as a replication or pooled estimate.

S8 Latency correction and release identity

The earlier manuscript/Supplement mismatch for the OCM-Asym CPU batch-1 value is closed by generating Table S15 directly from `paper_outputs/release_v1/inference_latency_protocol_summary.csv`. The authoritative median is 1.612 ms, not 1.167 ms. The former Supplementary Table S27 is superseded by the following generated exhibit; no manually transcribed timing value is authoritative.

S9 Analysis chronology

The timeline separates original training, result-informed reanalysis, external battery freezing, later artifact correction, and the present journal revision. Dates and clock times added for the full crossing are local Asia/Shanghai time unless a UTC suffix is printed. The 22 September entry records

Table S15: Repeated FD004 forward-pass timing. GPU measurements use 50 warm-up and 200 synchronized timed iterations; CPU measurements use 20 warm-up and 100 timed iterations. Pre-processing and host–device transfer are excluded. Values summarize five independently trained checkpoints.

Point	Params	Device	Batch	Median ms	Checkpoint range	Median [p10,p90]
OCM-Asym	54,155	CPU	1	1.612	[1.582, 1.999]	[1.389, 2.057]
OCM-Asym	54,155	GPU	1	1.886	[1.584, 2.317]	[1.484, 2.295]
OCM-Asym	54,155	GPU	128	1.990	[1.720, 2.159]	[1.603, 2.275]
RAST-GRU	87,568	CPU	1	1.658	[1.610, 1.723]	[1.547, 1.839]
RAST-GRU	87,568	GPU	1	1.836	[1.656, 2.076]	[1.511, 2.080]
RAST-GRU	87,568	GPU	128	1.868	[1.574, 2.665]	[1.440, 2.484]

that execution was already underway when the plan was frozen; it does not assert pre-execution registration. A later public release can repair accessibility but cannot change a retrospective analysis into prospective evidence.

Table S16: Complete analysis and release timeline. Later artifact repair does not change earlier design timing.

Date/period	Event	Source record	Evidence status
2026-07-04	C-MAPSS project scaffold and initial experiment design	local project records	precedes later governance analyses; no external timestamp
before	Common-protocol C-MAPSS training grid completed	analysis chronology	results later inspected; retrospective evidence
2026-07-20	Crossed seed-engine reanalysis and threshold diagnostics	analysis chronology	result-informed reanalysis; no retraining
2026-07-26	Normalization, perturbation, mask, and latency audits	analysis chronology	post-result sensitivity families
2026-07-26/27	Retrospective extension and release-governance revisions	round 10–19 response records	completed after earlier result access
2026-07-29 to 2026-08-08	Evidence snapshot	workflow state	local evidence snapshot; not a prospective protocol
2026-08-09	2026-08-09-r21 assembled		
2026-08-12 09:10:33Z	Battery protocol externally registered	external timestamp record	precedes battery training and test-label evaluation
after	Registered battery pair trained and evaluated	battery results audit	prospectively frozen worked execution; same investigator
2026-08-12 registration	Source provenance correction verified in v5.3.1	Git tag v5.3.1 and repository history	corrected source snapshot; does not alter evidence timing
2026-08-16	RESS major-revision exploratory analyses generated	major-revision evidence generator	post-result maintenance, equivalence, baseline, and target-definition sensitivities
2026-08-16	RESS submission object rebuilt	Git tag v5.4.0 and submission_identity.json	release synchronization and journal-format revision; no evidence-status upgrade
2026-08-16	v5.4.0-p1 local submission-layer patch assembled	submission patch record	abstract reduced to journal limit; manifest self-hash corrected; selected-grid table transcription repaired; stale unused macros removed; no scientific result changed

Date/period	Event	Source record	Evidence status
2026-08-17	v5.4.0-p2 local submission-compliance patch assembled	submission patch record	corresponding-author contact completed; formal OSF dataset citation added; vector Figure 5 added; PDF-first artwork and source-package bibliography resolution verified; no scientific result changed
2026-08-17	v5.4.0-p3 privacy-safe submission patch assembled	submission patch record	residential contact isolated to confidential Editorial Manager metadata; public manuscript/source scrubbed; declaration mapping and portable test contract repaired; no scientific result changed
2026-08-20	v5.4.0-p4 final migration-alignment patch assembled	submission patch record and migrated GitHub repository	migrated GitHub tag resolved to b48c7e302cb0...; repository URL/identity synchronized; figure/table text citations and evidence-manifest chronology repaired; no scientific result changed
2026-08-20	v5.4.0-p5 submission-documentation and compatibility patch assembled	submission patch record	disclosure wording and document metadata normalized; no scientific result changed
2026-09-12	v5.4.0-p6-readability readability revision assembled	submission patch record	abstract, introduction, section openings, discussion, and conclusion restructured for broad reliability-engineering readability; scientific result changed: no
2026-09-12	v5.4.0-p7-editorial-polish editorial-polish revision assembled	submission patch record	title, abstract, front-loaded terminology, section headings, captions, and submission-facing descriptions polished for editor readability; scientific result changed: no
2026-09-12	v5.4.0-p8-reviewer-risk-reduction scientific-narrative revision assembled	submission patch record	scientific contribution order, research questions, related-work emphasis, results/discussion flow, and submission-facing descriptions restructured to foreground RUL evaluation robustness and bounded decision relevance; scientific result changed: no

Date/period	Event	Source record	Evidence status
2026-09-12	v5.4.0-p9-reviewer-hardening reviewer-hardening revision assembled	submission patch record	title reframed as protocol sensitivity; scientific contributions reduced to three; C-MAPSS rationale clarified; endpoint-role, N-CMAPSS, and reproducibility detail moved to Supplement; scientific result changed: no
2026-09-22	Full 5-by-10 execution was underway before analysis plan and Amendment 01 were frozen	local launcher, plan, Amendment 01, and run manifest	pre-aggregation analysis lock; not pre-execution preregistration
2026-09-25 18:40	Full 400-cell run manifest complete; configuration and artifact QA passed	canonical manifest and per-run outputs	400 completed cells; all runs stopped under the declared patience rule
2026-09-25 19:06	Frozen finite-design aggregation and pairwise rank-support tables generated	aggregation provenance and output hashes	400 runs and 200 matched split-stream cells in total; descriptive only
2026-09-25 19:46	Observed-prefix dynamic policy grid completed	policy protocol, summary, paired comparison, and engine-level outputs	post-result exploratory grid; no field or policy-benefit claim

S10 Pre-outcome-frozen battery study details

Discharge observations are indexed by $j = 0, \dots, J_i$. The trained target is $J_i - j$; $J_i + 1$ is the observed discharge count. Four held-out batteries are the independent units. Five training streams are repeated fitting variations, and eligible windows are metric denominators rather than independent replicates.

Table S17: Prospectively frozen NASA battery evaluation. Values are the mean and observed range over five registered training streams. Differences are Battery-GRU-Asym minus Battery-GRU-Core. Counts describe stream-level signs and are not inferential probabilities.

Configuration/contrast	Macro RMSE (cycles)	LPR _{10%}	Direction support
Battery-GRU-Core	30.051 [21.299, 33.687]	0.8610 [0.5172, 1.0000]	5 streams
Battery-GRU-Asym	32.596 [22.166, 36.776]	0.7833 [0.5909, 1.0000]	5 streams
Asym – Core	+2.546	-0.0777	RMSE 0/0/5; LPR 2/2/1

Direction support is negative/zero/positive. The four test batteries contribute equally within every stream. The registered rule classifies the joint result as an observed accuracy–late-overprediction trade-off.

Table S18: Battery-level support averaged over five registered streams. Each battery is an independent evaluation unit; window counts are denominators, not replicates.

Battery	Eligible	Core RMSE	Asym RMSE	Δ RMSE	Core LPR	Asym LPR	Δ LPR
B0007	17	58.086	62.041	+3.955	0.8471	1.0000	+0.1529
B0044	12	27.490	30.696	+3.206	0.8333	0.5333	-0.3000
B0048	8	11.517	11.818	+0.300	1.0000	1.0000	+0.0000
B0055	11	23.111	25.831	+2.720	0.7636	0.6000	-0.1636

Table S19: Post-registration exploratory battery baselines under global training-only normalization.

Estimator	Macro RMSE	Macro LPR _{10%}	Pooled late/eligible	Pooled frequency
training-life-prior	23.720	0.2500	8.0/48.0	0.1667
capacity-index-linear	25.998	0.8824	40.0/48.0	0.8333
window-summary-ridge	12.711	0.9773	47.0/48.0	0.9792
Battery-GRU-Core (five-stream mean)	30.051	0.8610	40.8/48.0	0.8500
Battery-GRU-Asym (five-stream mean)	32.596	0.7833	38.0/48.0	0.7917

Notes: The three simple estimators were added after registration and do not enter the registered hypothesis or wording rule. GRU rows are five-training-stream means. LPR is an equal-battery macro average; pooled counts use 48 eligible windows per stream for the registered models and must not be reverse-calculated from macro LPR.

Table S20: Battery target-denominator definition sensitivity at the primary $q = 0.10$, $\epsilon = 0$ setting.

Denominator	Δ macro RMSE	Δ macro LPR	Core/Asym pooled events	Eligible windows
J_i	+2.5455	-0.0777	204/190	240
$J_i + 1$	+2.5455	-0.0777	204/190	240

Notes: J_i is the final zero-based discharge index and $J_i + 1$ is the observed discharge-count denominator. The trained target and predictions remain $J_i - j$; only the life-fraction denominator changes. RMSE is therefore invariant. Event counts are pooled over four batteries and five streams, whereas LPR is first averaged equally across batteries and then across streams.

The registered global-normalization LPR is an equal-battery macro average. Pooled event counts over 48 eligible windows need not equal macro LPR multiplied by 48. Over the complete registered (q, ϵ) grid and both normalization rules, changing the life-fraction denominator from J_i to $J_i + 1$ shifts the Asym-minus-Core macro-LPR contrast by at most 0.0625 in absolute value; the primary cell is unchanged.

S11 Independent comparator-study status

An executable study-ready package is located at `studies/six_link_comparator_study/`. It specifies a randomized blinded-label comparison, a target of at least 12 independent raters, balanced packet assignment, a frozen defect key, and rater-level outcomes for defect detection, false positives, claim-strength agreement, completion time, and burden. At this release the response file contains zero participants. The author is not counted as an independent rater, generative-AI reviewers do not satisfy that criterion, and recruitment is blocked pending a human-participant/ethics determination. The only permitted current conclusion is that two author-run worked records and a future comparator protocol exist; comparative efficacy is untested. Table S21 records the readiness state and resulting claim boundary.

Table S21: Comparator-study readiness and claim boundary.

Object	Current state	Consequence for this article
Protocol and construct map	Prepared after current result inspection	May define a future study; no prospective status here
Assignment and response schemas	Present and validated	Demonstrates executability of data capture only
Independent rater responses	0	No effect size, agreement, timing, or burden result may be reported
Ethics determination	Not yet obtained	No recruitment is authorized by this package
Current scaffold evidence	Two same-author worked cases	Report as traceability support, not an effective intervention

S12 Reproducibility checks and artifact record

Reproducibility checks are reported at three distinct levels. *Stored-result verification* checks machine-readable outputs, row counts, manifests, and hashes without retraining. *Exhibit regeneration* rebuilds reader-facing tables and figures from declared source rows. *Full or sentinel retraining* reruns a declared training path and is held to a prediction-level criterion. Passing one level does not imply passing the others, and none changes the original timing status of an analysis.

For C-MAPSS, the release record reports a clean stored-result verification and one same-stack FD001 seed-42 sentinel refit that exactly reproduced all 100 RAST-GRU and OCM-Asym predictions, including selected epochs and event summaries. The compact submission supports manifest, hash, schema, and stored-output checks but does not contain the full checkpoint/prediction tree needed to rerun that sentinel refit. The complete post-study evidence archive is available at OSF <https://osf.io/u76ze/>, DOI 10.17605/OSF.IO/U76ZE; a public source and verified-output layer is available at <https://github.com/Ethan0infinity/rul-benchmark-protocol-audit>. These records document bounded reproduction paths, not investigator-independent reproducibility or prospective C-MAPSS evidence.

For the battery study, the externally timestamped protocol ZIP and the locally executed protocol ZIP are byte-identical, with SHA-256 `f192bb6bdd6188742bb7da4bf17d3fec164e5e2b5ccd87eb272dbb4f1dbd11fd`. Prediction, label, result, history, environment-deviation, and execution-hash records are retained separately after execution. The post-execution evidence object is included unchanged in the public OSF archive. These checks support two bounded statements: the registered protocol preceded outcome evaluation, and the completed evidence is inspectable. They do not provide investigator independence or show that the traceability scaffold improves reproducibility in other studies.

Selected claim-to-artifact navigation appears below. Complete paths, hashes, commands, and generator identities remain in the release records and OSF evidence archive.

S13 Artifact navigation and identity contract

Table S22 lists the core machine-readable sources used by this concise submission.

Table S22: Core machine-readable sources for the concise submission.

Claim family	Source	Generator or verifier
Fixed C-MAPSS effects	<code>paper_outputs/release_v1/primary_seed_level_effects.csv</code>	release-v1 exhibit generators
Full split-stream crossing	<code>paper_outputs/full_5x10_crossing/analysis/full_5x10_paired_contrasts.csv</code>	<code>scripts/aggregate_full_5x10_crossing.py</code>
Observed-prefix policy	<code>paper_outputs/full_5x10_crossing/dynamic_policy/dynamic_policy_paired_model_comparison.csv</code>	<code>scripts/run_dynamic_maintenance_policy.py</code>
Maintenance decisions	<code>paper_outputs/ress_major_revision/maintenance_decision_cell_level.csv</code>	stored-output verification; complete inputs required for regeneration
Candidate magnitude bands	<code>paper_outputs/ress_major_revision/*practical_equivalence.csv</code>	stored-output verification; complete inputs required for regeneration
Battery baselines and J_i audit	<code>paper_outputs/ress_major_revision/battery_*.csv</code>	hashes verified locally; complete inputs required for regeneration
Comparator study	<code>studies/six_link_comparator_study/</code>	<code>scripts/prepare_comparator_study.py --check-only</code>
Submission identity	<code>submission_identity.json</code>	release build script and independent SHA-256 check

The release identity uses one upstream annotated tag plus an external local-patch build record. The compact major-revision manifest inventories stored outputs without self-hashing; its generator requires the complete prediction and prepared-data inputs listed in that manifest for regeneration. A Git commit cannot contain its own cryptographic identifier without a self-reference paradox;

therefore the tagged source contains the build logic and release tag, while the regenerated submission object records the resolved tag commit, tagged tree, source-content commit, final PDF hashes, archive hash, and timestamp in `submission_identity.json`. This contract avoids presenting a prior commit, stale PDF, or mutable branch name as the identity of the submitted object.

The 5-by-10 crossing and observed-prefix policy outputs summarized here are included in the version-named OSF evidence archive listed above. The OSF project DOI identifies the project record; no separate version DOI is assigned to this file. GitHub source and compiled-document identities are recorded separately so that the evidence ZIP, source tree, and PDFs are not conflated.

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